

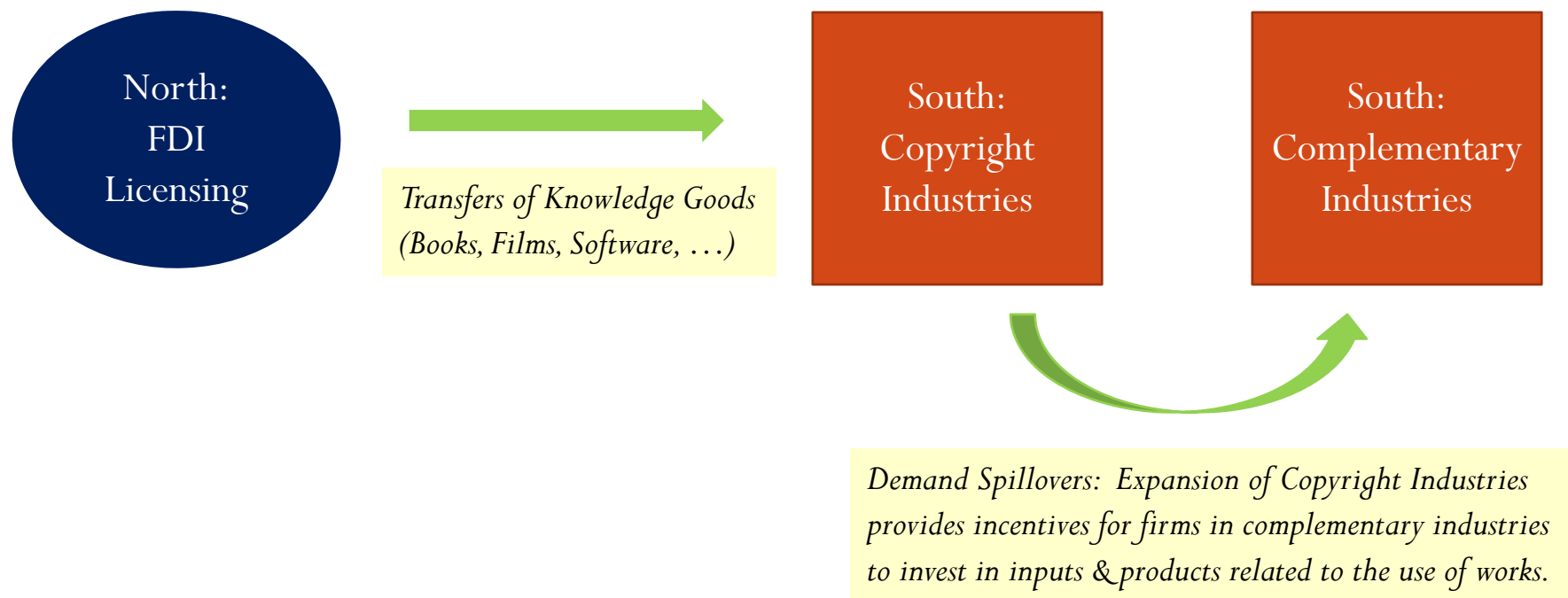
Copyright and Economic Development: Evidence from Multinational Investments

Walter G. Park
American University
November 2014

The statistical analysis of firm level data on U.S. multinational companies and their foreign affiliates was conducted at the Bureau of Economic Analysis, United States Department of Commerce, under arrangements that maintain legal confidentiality requirements. Views expressed in this paper are those of the authors and do not necessarily reflect the official positions of the US Department of Commerce.

Conceptual Framework

- Two Sources of 'Knowledge Capital' in Developing Countries: (i) Inward Technology Transfer (Licensing, Foreign Direct Investment (FDI)) ; (ii) Local Investments (in R&D and Physical Capital)



- Empirical Goals are to estimate the impacts of
 - Copyright Protection on Inward Technology Transfer
 - Market Size of Copyright Industries on Investments in Complementary Industries
 - Inward technology and local investments on local productivity

Nature of Dataset

- 1,223 Parent Firms
- Each has at least one or more affiliates in the
 - North (23 Developed countries) 69% of sample
 - South (65 Developing Countries) 31%
- 3 Main Industries
 - Copyright-related (Information) 24% of sample
 - Complementary (High-technology) 60%
 - Professional & Scientific Services 16%
- Annual Data: 1989 - 2011

Characteristics of Firms in the ‘South’

Mean 1989-2011	Copyright-related Industries	Complementary Industries	Professional & Scientific Industries
Parent R&D/Sales	1.72%	9.11%	8.36%
Affiliate R&D/Sales	0.93%	0.75%	0.85%
Parent Capital/Labor Ratio	58.9	45.8	16.1
Affiliate Capital/Labor Ratio	15.4	15.7	4.1
Parent Productivity	114	107.6	142
Affiliate Productivity	25.3	28.4	18.9

Notes: Capital/Labor is in thousands of real 2005 U.S. dollars per worker, Productivity is defined as Gross Product (thousands of real 2005 U.S. dollars) per worker.

Distribution of Technology Transfer to the 'South'

Naics code	Copyright-Related Industries	FDI Stock	Licensing Receipts
5111	Newspaper, periodical, book, and database publishers	41.5%	2.58%
5112	Software publishers	9.5%	9.14%
5121	Motion picture and video industries	45.0%	82.06%
5122	Sound recording industries	2.1%	0.07%
5152	Cable networks and program distribution	0.0%	3.03%
5172	Wireless telecommunications carriers, non satellite	0.2%	0.00%
5191	Other Information Services	1.8%	0.00%
Naics code	Complementary Industries	FDI Stock	Licensing Receipts
3231	Printing and related support activities	1.8%	0.27%
3341	Computer and peripheral equipment manufacturing	24.8%	24.88%
3342	Communications equipment manufacturing	27.9%	69.00%
3343	Audio and video equipment manufacturing	0.3%	
3344	Semiconductors and other electronic components manufacturing	42.6%	5.70%
3345	Navigational, measuring, electromedical, and control instruments	0.5%	0.09%
3351	Electric lighting equipment manufacturing	0.1%	0.07%
3359	Other Electrical Equip	2.0%	0.00%
Naics code	Professional & Scientific	FDI Stock	Licensing Receipts
5411	Legal services	9.8%	0.00%
5412	Accounting, tax preparation, bookkeeping, and payroll services	0.8%	
5413	Architectural, engineering, and related services	2.2%	0.00%
5415	Computer systems design and related services	36.0%	71.01%
5416	Management, scientific, and technical consulting services	25.0%	28.96%
5417	Scientific research and development services	2.0%	0.00%
5418	Advertising and related services	22.3%	
5419	Other professional, scientific, and technical services	1.8%	0.03%

Mean FDI (Licensing) in copyright industries is 20% (32%) of that in the complementary industries

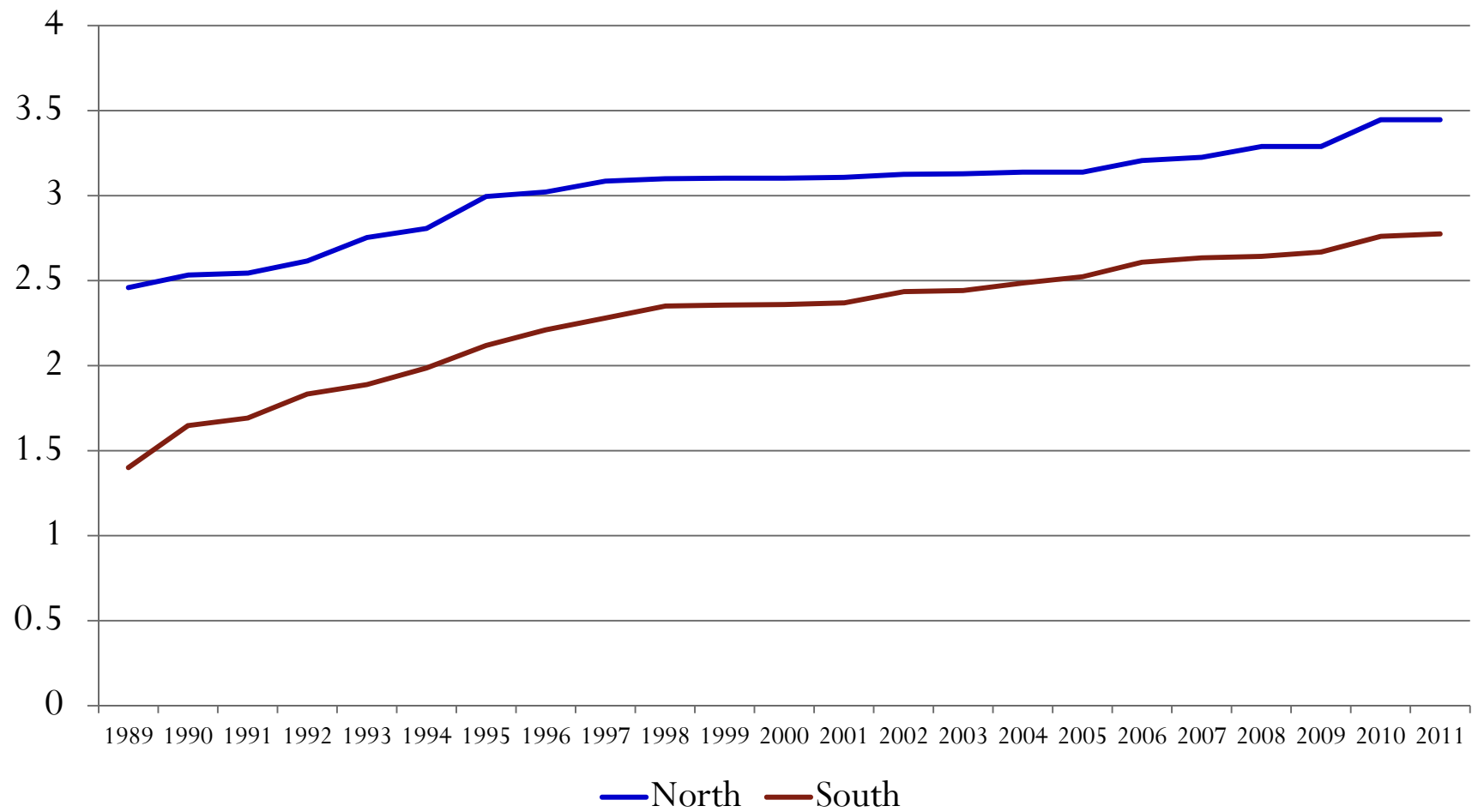
Distribution of Licensing by type of Intellectual Property in Developing Countries

Naics code	Sector	Industrial Processes	Books, Tapes	Trade marks	Films	Performances	Broadcasting	Software	Franchise	Other
5111	Newspaper, periodical, book, and database publishers	14.57%	47.74%	29.03%	3.90%	4.47%			0.30%	
5112	Software publishers	0.01%		4.77%				95.22%		
5121	Motion picture and video industries	0.04%	2.44%	8.84%	41.40%	36.62%	9.42%	0.59%		0.66%
5122	Sound recording industries		100.00%							
5151	Radio and television broadcasting		4.95%	24.21%	0.01%	56.95%	0.62%	1.13%	12.15%	
5152	Cable networks and program distribution		0.99%			99.01%				
5191	Other Information Services									
3231	Printing and related support activities	1.25%		93.76%					4.99%	
3341	Computer and peripheral equipment manufacturing	90.35%		0.40%				9.25%		
3342	Communications equipment manufacturing	89.04%	0.24%	0.03%		10.68%		0.01%		
3344	Semiconductors and other electronic components manufacturing	99.48%						0.52%		
3345	Navigational, measuring, electromedical, and control instruments manufacturing	97.62%						2.38%		
3346	Manufacturing and reproducing magnetic and optical media							100.00%		
5415	Computer systems design and related services	0.00%						99.87%	0.13%	
5416	Management, scientific, and technical consulting services		0.08%						99.92%	
5419	Other professional, scientific, and technical services			100.00%						

Index of Copyright Protection

- Park (2005) and Reynolds (2003), updated
- Index covers 120 countries from 1965 – 2013 (annually, where available).
- Four Components (measuring strength of protection)
 - Duration (Copyrights and Related Rights)
 - Usage (Private Use, Collective Licensing)
 - Enforcement (Injunction, Criminal Penalties, Anti-circumvention)
 - International Agreements/Treaties/Conventions
- Overall Index varies from 0 – 4.
- Source of Information: National Laws and Commentaries

... Trends



Post-reform Impacts

Ratio of Variables Post-Reform/Pre-Reform in the 'SOUTH':

Industry Group:	FDI Stock	Unaffiliated Licensing	R&D by Affiliates	Capital Stock of Affiliates	Employment of Affiliates	Value Added of Affiliates
Copyright-related	1.91	1.72	21.92	2.10	2.39	1.70
Complementary	0.79	3.69	2.10	1.11	0.91	1.06
Professional & Scientific	1.41	0.64	3.15	1.94	3.24	2.34

Reform Dummy Variable = 1 for years after a major reform
= 0 otherwise

For example, $\left(\frac{\text{Mean FDI Stock in the Copyright Industries of the South after the year of major reform}}{\text{Mean FDI Stock in the Copyright Industries of the South before the year of major reform}} \right) = 1.91$

Post-reform Impacts

Other Reforms

Industry Group:	FDI Stock	Unaffiliated Licensing	R&D by Affiliates	Capital Stock of Affiliates	Employment of Affiliates	Value Added of Affiliates
Post-TRIPS						
Copyright-related	2.24	1.94	8.62	3.65	2.76	1.46
Complementary	1.39	1.42	1.90	1.52	1.12	1.50
Professional & Scientific	1.97	0.20	1.39	2.79	3.38	2.68
Post-WIPO COPYRIGHT TREATY						
Copyright-related	1.76	1.26	0.49	0.75	1.03	0.87
Complementary	0.94	1.09	1.39	1.33	1.03	1.10
Professional & Scientific	1.26	0.49	2.94	1.34	1.18	1.22
Post-WIPO PERFORMANCES & PHONOGRAMS TREATY						
Copyright-related	1.70	1.18	0.53	0.82	1.11	0.91
Complementary	0.93	0.58	1.26	1.28	1.08	1.09
Professional & Scientific	1.07	0.42	3.16	1.43	1.26	1.18

Note: Each entry shows the ratio of a variable post-agreement (treaty) to its pre-agreement (treaty) value

Tech Transfer Regressions, using 'SOUTH' sample:

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DEPENDENT VARIABLES:	ln (Licensing)	ln (Licensing)	ln (Licensing)	ln (Licensing)	ln(FDI)	ln(FDI)	ln(FDI)	ln(FDI)
Constant	120.623*** (19.301)	53.940 (94.765)	53.224 (90.595)	27.531 (96.854)	32.304 (39.120)	134.082*** (47.054)	153.722*** (47.644)	175.680*** (48.721)
ln (GDP)	0.699** (0.338)	2.135*** (0.368)	2.186*** (0.368)	2.156*** (0.364)	1.439*** (0.533)	1.421** (0.589)	1.439** (0.589)	1.500** (0.591)
ln (Copyright Index)	0.825*** (0.181)	0.031 (0.189)	-0.210 (0.222)	-0.142 (0.198)	0.985*** (0.282)	0.701*** (0.207)	0.555*** (0.203)	1.040*** (0.231)
ln (Copyright Index) x Market Rank			0.097** (0.042)				0.079** (0.031)	
ln (Copyright Index) x Market Rank 1st Quartile				-0.330 (0.208)				-0.419*** (0.115)
ln (Copyright Index) x Market Rank 3rd Quartile				0.596*** (0.169)				-0.456*** (0.168)
ln (Copyright Index) x Market Rank 4th Quartile				0.271*** (0.076)				
ln (Capital/Labor)	0.068*** (0.022)	0.061*** (0.021)	0.058*** (0.021)	0.064*** (0.021)	0.124*** (0.043)	0.123*** (0.042)	0.126*** (0.043)	0.127*** (0.043)
Tax Rate (Destination)	-0.132* (0.070)	-0.059 (0.058)	-0.058 (0.059)	-0.053 (0.058)	-0.000 (0.001)	0.002 (0.001)	0.002 (0.001)	0.002 (0.001)
Year Fixed Effects	Not Included	Included	Included	Included	Not Included	Included	Included	Included
Country Fixed Effects	Included	Included	Included	Included	Included	Included	Included	Included
Industry Fixed Effects	Included	Included	Included	Included	Included	Included	Included	Included
Observations	4,818	4,818	4,818	4,818	5,014	5,014	5,014	5,014

Notes: Robust standard errors in parentheses, clustered by country and year. *** indicates significance at 1% level, ** 5%, and * 10% level

Investment Regressions, using 'SOUTH' sample:

SECTOR:	Complementary Industries			Professional & Scientific		
	(1)	(2)	(3)	(4)	(5)	(6)
DEPENDENT VARIABLES:	ln (Affiliate R&D)	ln (Affiliate Capital)	ln (Affiliate Sales)	ln (Affiliate R&D)	ln (Affiliate Capital)	ln (Affiliate Sales)
Constant	-249.874*** (43.433)	81.175*** (26.326)	-39.945 (32.320)	-73.121* (38.153)	-26.889 (39.081)	-149.869*** (41.585)
ln (Affiliate Sales)	0.334*** (0.031)			0.125*** (0.029)		
ln (GDP)	-1.752*** (0.545)	1.276*** (0.352)	1.764*** (0.347)	-0.945** (0.475)	1.102* (0.610)	0.712 (0.487)
ln (Copyright Index)	0.394*** (0.125)	0.058 (0.074)	0.278*** (0.095)	0.348*** (0.133)	0.268* (0.139)	-0.145 (0.209)
ln (Market Size of Copyright Industries)	-0.066 (0.043)	0.113*** (0.030)	0.071*** (0.027)	0.021 (0.046)	0.117** (0.050)	0.164*** (0.039)
ln (Capital/Labor)	-0.093 (0.060)	0.388*** (0.071)	0.242*** (0.040)	0.049 (0.042)	0.111** (0.052)	0.205*** (0.040)
Tax Rate (Destination)	0.007*** (0.002)	0.001 (0.001)	0.001 (0.002)	-0.002 (0.001)	-0.001 (0.001)	0.001 (0.002)
Year Fixed Effects	Included	Included	Included	Included	Included	Included
Country Fixed Effects	Included	Included	Included	Included	Included	Included
Industry Fixed Effects	Included	Included	Included	Included	Included	Included
Observations	7,787	7,789	7,790	2,477	2,479	2,479

Notes: Robust standard errors in parentheses, clustered by country and year. *** indicates significance at 1% level, ** 5%, and * 10% level

National-Level Regressions:

Dependent Variable: ln (Patent Priority Filings by Inventor Country)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Sample:	SOUTH	SOUTH	SOUTH	SOUTH	SOUTH	SOUTH	SOUTH
Constant	0.394* (0.229)	-0.793 (0.595)	9.851*** (0.246)	0.422* (0.227)	-1.528*** (0.469)	-1.006*** (0.333)	9.901*** (0.250)
ln (Unaffiliated Licensing in Complementary Industries)	0.053*** (0.013)		0.060*** (0.015)				0.063*** (0.015)
ln (Unaffiliated Licensing in Copyright Industries)		0.019 (0.024)	-0.009 (0.026)				-0.000 (0.027)
ln (FDI Stock in Complementary Industries)				0.024*** (0.009)		0.005 (0.011)	-0.007 (0.011)
ln (FDI Stock in in Copyright Industries)					-0.009 (0.012)	-0.011 (0.011)	-0.017 (0.011)
Year Fixed Effects	Included	Included	Included	Included	Included	Included	Included
Country Fixed Effects	Included	Included	Included	Included	Included	Included	Included
Observations	1,234	756	745	1,257	760	748	744

Robust standard errors in parentheses. *** indicates significance at 1% level, ** 5%, and * 10% level

Licensing refers to 'technologies or know-how' **received by** the industry.

$$\text{Motivation : } A = \frac{Y}{F(K, L)} = A(\text{Inventions, ...})$$

- Other results

By 'North' Sample

Robustness Tests

- Selected References

Ann, S., Hall, B. and Lee, K. (2014) *Intellectual Property for Economic Development*, Edward Elgar.

Bilir, K.L. (2014), "Patent Laws, Product Lifecycle Lengths, and Multinational Activity," *American Economic Review*, Vol. 104, No. 7.

Geller, P. and Bently, L. (2013), *International Copyright Law and Practice*, Matthew Bender.

Mataloni, R. (1995), "A Guide to BEA Statistics on U.S. Multinationals," *Survey of Current Business*

Park, W. G. (2005), "Do Intellectual Property Rights Stimulate R&D and Productivity Growth? Evidence from Cross-National and Manufacturing Industry Data," in J. Putnam (ed.) *Intellectual Property Rights and Innovation in the Knowledge-Based Economy*, Industry Canada, Ottawa.

Reynolds, T. (2003), *Quantifying the Evolution of Copyright and Trademark Law*, PhD thesis, American University.